



Experimental and modeling efforts towards a better understanding of the high-temperature combustion kinetics of C₃–C₅ ethyl esters



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ABSTRACT

Kinetic investigations into small ethyl esters lay the foundation for a good understanding of combustion properties, particularly those brought about by the ethyl ester functional group, for biodiesels in forms of fatty acid ethyl esters. A systematical study was conducted in this work, aimed at better interpreting the high-temperature reaction schemes for the simple C₃–C₅ ethyl esters. Quantitative information for crucial intermediates closely related to fuel consumptions was probed in the low-pressure premixed flame of each ester with a molecular beam mass spectrometer. Efforts based on theoretical calculations and extensive literature reviews were devoted to developing a kinetic model which can well predict the current experimental flame structures as well as speciation measurements under pyrolysis conditions reported in literature (Ren et al. (2014)). Model analyses in combination with experimental results helped in revealing the significant effects of specific reactions on the fates of key intermediates in ethyl esters combustion. The competitions between eliminations producing ethylene and acids and hydrogen abstractions by H atoms account for fuel consumptions under flame conditions, but the relative importance of the two reaction categories vary among the three flames. During the shock tube pyrolysis, acids are rapidly produced from fuel decompositions, but consumed in much longer periods, resulting in species formation and enriching the radical pools.

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1. Introduction

Biodiesels have been proposed and widely applied as alternative fuels to relieve the negative environmental consequences of burning fossil fuels [1–3]. Combustion kinetics of biodiesels, usually mixtures of C₁₂–C₂₂ mono-methyl and -ethyl esters, have been pursued to gain insights into the mechanisms for energy conversion and pollutants formation [4,5]. Specific kinetic characteristics related to ester functional groups can be better revealed through relevant investigations on small esters [6]. Great attention has been paid to the combustion chemistry of small methyl esters [7–10], basically because methanol is preferred in the transesterification producing fatty acid methyl ester (FAME) biodiesels with relatively low costs. Nevertheless, the extent of renewability can be further enhanced to produce biodiesels in the form of fatty acid ethyl esters (FAEEs) with bioethanol which also possesses merits of lower toxicity and corrosivity compared to methanol. Combustion kinetics of small ethyl esters in most previous studies was addressed

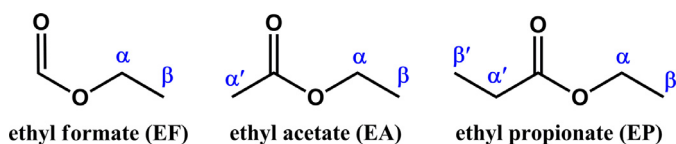
in comparisons with that of relevant methyl esters. Methyl butanoate and ethyl propionate (C₅H₁₀O₂) are the most studied isomer pair, whose fuel consumption pathways and species formations have been compared under different conditions, including the shock tube ignition [11], the rapid compression machine (RCM) oxidation [12] and low-pressure premixed flames [13]. By comparing ignition behaviors for small methyl and ethyl esters of the same alknoic acids, Akih-Kumgeh and Bergthorson [14] reported the higher reactivity of ethyl esters than their methyl ester counterparts. All aforementioned studies emphasized structural effects on combustion characteristics of small esters, while systematical kinetic investigations targeted at ethyl esters are still rare. In a shock tube study by Ren et al. [15], concentration time histories of CO₂, CO, H₂O were recorded during the pyrolysis of C₃–C₅ ethyl esters, namely ethyl formate (EF), ethyl acetate (EA) and ethyl propionate (EP), as presented in Scheme 1. Modeling efforts [15] were made to reproduce the measurements and elucidate important reaction pathways, but large discrepancies existed between the experimental and simulated results in the case of EP, reflecting that more theoretical works are desired to better establish the fuel decay patterns.

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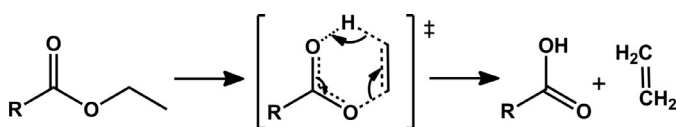
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Table 1
Detailed conditions for C₃–C₅ ethyl ester flames in this study.

Flame	Ethyl formate (EF)	Ethyl acetate (EA)	Ethyl propionate (EP)
Pressure (Torr)	30.0	30.0	30.0
Equivalence ratio	1.50	1.50	1.50
Mass flow rate (g/cm ² /s)	0.00271	0.00272	0.00273
Total flow rate (SLM)	2.438	2.438	2.438
Fuel flow rate (SLM)	0.366	0.281	0.229
O ₂ flow rate (SLM)	0.853	0.938	0.990
Ar flow rate (SLM)	1.219	1.219	1.219



Scheme 1. Chemical structures for C₃–C₅ ethyl esters investigated in the shock tube study by Ren et al. [15] as well as the present work.



Scheme 2. The intra-molecular elimination reactions dominating unimolecular decompositions of ethyl esters. R is H or an alkyl moiety.

In addition to the practical significance that kinetic studies for small ethyl esters are meaningful to controllable applications of FAEE biodiesels, reacting systems of small ethyl esters are found attractive because they allow investigations into specific reactions, like intra-molecular eliminations. The hydrogen atom transfer from the β -carbon to the carbonyl oxygen, as shown in Scheme 2, has been proven to be the overwhelmingly dominant channel for ethyl esters thermal decomposition, producing ethylene (C₂H₄) and corresponding acids. Small acids with one to three carbon atoms are important combustion intermediates [16], but limited is known about their combustion kinetics because experimental investigations are difficult due to the high viscosity and corrosivity of acids. Pyrolysis and combustion of ethyl esters potentially provide abundant acids whose consumption pathways can thus be shed light on.

In the present work, experimental and modeling efforts were combined to explore the high-temperature kinetics of C₃–C₅ ethyl esters. Specifically, premixed flames respectively fueled by EF, EA and EP were experimentally studied under almost identical conditions to facilitate comparisons among these flames. Crucial intermediates such as acids were well identified and quantified with the technique of molecular beam mass spectrometry (MBMS). Based on high-level theoretical calculations and extensive reviews on recent literatures, a new kinetic model for C₃–C₅ ethyl esters high-temperature combustion was proposed. Speciation measurements in the current study as well as those under pyrolysis conditions reported by Ren et al. [15] were used for comprehensive model validations. Emphasis was put on elucidating the roles played by specific reactions and their temperature and pressure dependence, which will determine the fates of fuels and key intermediates, thus altering the species pool and the reactivity of the reaction systems under different conditions.

2. Experimental method

The flame structure measurements were accomplished at the National Synchrotron Radiation Laboratory in Hefei, China. The apparatus has been described in detail by Qi et al. [17,18], so here

only a brief summary will be provided. It mainly consists of a low-pressure flame chamber, a molecular beam sampling system and a time-of-flight mass spectrometer (TOFMS) using tunable vacuum-ultraviolet (VUV) synchrotron for photoionization. Fuel rich ($\phi=1.50$) flames respectively fueled by EF, EA and EP were stabilized on a 60-mm-diameter McKenna burner at the pressure of 30.0 Torr to maximum the flame thickness and stand-off distance while maintaining the flame stability. The total flow rate of the gas mixture was kept at 2.438 standard liter per minute (SLM) with an argon (Ar) dilution ratio of 0.50 for each flame. Almost equivalent flame conditions can be seen from Table 1 for the three flames, among which potential comparisons could therefore be made. The gas flow rates of oxygen (O₂) and Ar were separately controlled by mass flow controllers while the liquid fuels into the vaporizer were metered by a chromatographic injection pump. Errors for all flow rates were estimated within 2% of the target values.

The flame gases were sampled along the flame axis by a quartz cone, with a lot of reactive species preserved in the forming molecular beam which was photo-ionized by the VUV light. The resulting ions were subsequent analyzed by a TOFMS with the mass resolution ($m/\Delta m$) up to 3000 so that species with the same mass number but different molecular weights could be separated over the mass range covered by current measurements. When fixing the burner at specific positions while scanning the photon energies, photon ionization efficiency (PIE) curves were obtained for species identification [18]. Acids produced from the six-membered eliminations (see Scheme 2) are expected to be crucial intermediates in the current flames, so PIE curves for CH₂O₂ in the EF flame, C₂H₄O₂ in the EA flame and C₃H₆O₃ in the EP flame are demonstrated in Fig. 1, respectively. The presence of acids in corresponding flames can be confirmed by the thresholds representing the ionization energies (IEs) of C₁–C₃ acids as well as the shapes of measured PIE curves. But interferences from ester isomers or the fuel fragmentation existed in measurements for CH₃COOH and C₂H₅COOH in the EA and EP flames, respectively, according to the comparisons between PIE measurements and weighed photon-ionization cross sections (PICSSs) [19–21] for species with the same formulas. Thus burner scans were performed at selected photon energies to effectively distinguish isomers while excluding the influences from fuel fragmentations. For instance, the photon energy was fixed at 10.80 eV and 10.50 eV to, respectively, quantify CH₃COOH in the EA flame and C₂H₅COOH in the EP flame. The integrated signals for different species, as functions of the distance from the burner, were converted to mole fraction profiles by following the procedure proposed by Cool et al. [22].

The temperature profiles were measured with a 0.1-mm-diameter Pt-6%Rh/Pt-30%Rh thermocouple coated with Y₂O₃-BeO ceramic to reduce catalytic effects. And the measured temperatures were corrected by considering both the radiative heat loss [23] and cooling effects [24] of the sampling nozzle. The uncertainty for the temperatures was evaluated to be within 100 K. For species mole fraction profiles, the accuracy of the “distance from the burner” was estimated to be within 1 mm regarding the probe position. Experimental uncertainties for major species fractions could be

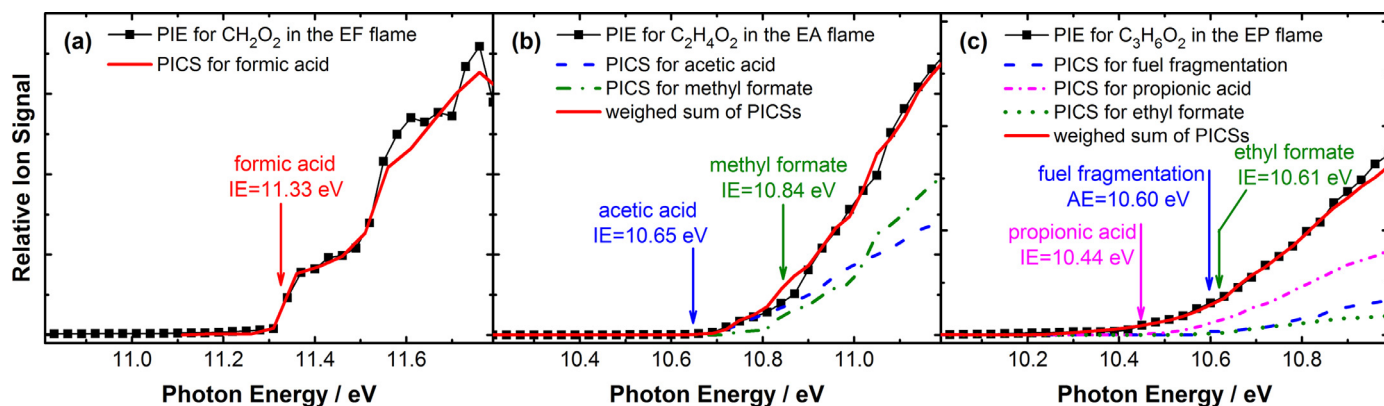


Fig. 1. The PIE curves measured at 3 mm away from the burner surface for acids and their isomers in corresponding flames. The PICSSs reported in literature (CH_2O_2 : [19] for formic acid; $\text{C}_2\text{H}_4\text{O}_2$: [21] for acetic acid, [20] for methyl formate; $\text{C}_3\text{H}_6\text{O}_2$: [21] for propionic acid, [20] for ethyl formate and [20] for ethyl propionate) were weighed to match separate PIE curves for the species identification.

within 20%, while comprehensive factors were considered to evaluate the uncertainties for intermediates concentrations. The uncertainty factors range from 30% to 70% for stable intermediates with measured PICSSs, while increasing to as large as 2 to 4 for radicals and species with estimated PICSSs. The uncertainty factors for peak concentrations of separate intermediates are provided in Table S1 in the Supplemental Material, together with the PICSSs and mass discrimination (MD) factors used in the quantification. Uncertainty factors for all species are larger in the region near the burner (0–1 mm) where perturbation from the sampling probe could be more severe [25].

All flame structure measurements in this work are provided in the Supplemental Material.

3. Theoretical calculation and kinetic modeling

In several previous kinetic studies related to small ethyl esters [14,15,26], the well-known kinetic models by Westbrook et al. [6,13] were adopted for simulations of experimental results and relevant analyses. And modifications to rate constants of specific reactions were proposed to better match the species concentrations under pyrolysis conditions [15]. In this work, we restarted from the Westbrook models [6,13] making updates based on theoretical calculations and recent literature reviews [9,10,27–38]. For the C_0 – C_4 core mechanism, the well-validated AramcoMech 1.3 [39] was adopted, whose earlier versions [40,41] were employed in Westbrook models [6,13]. Details for crucial reaction categories closely related to fuel consumptions will be addressed below. Table 2 lists the updated reactions in the primary mechanisms for ethyl ester fuels (reactions in Sections 3.1–3.3), with rate coefficients provided in the form of $k = A \cdot T^n \cdot \exp(-E_a/RT)$. While the sub-mechanisms for acids (reactions in Section 3.4) can be found in Table S2 in the Supplemental Material. To facilitate potential global sensitivity and uncertainty analyses for model optimizations in the future, herein uncertainty factors (*UFs*) for kinetic parameters in the fuel sub-mechanisms are recommended. For rate constants originated from literatures [27–29,36,38] with *UFs* also provided, these factors can be taken directly. And for rate coefficients theoretically derived in this work, a uniform *UF* of 3 is assigned. *UFs* ranging from 5 to 6 can be expected for kinetic parameters obtained via rate strategies or analogies. While for estimated rate constants, *UFs* as large as 10 are evaluated.

3.1. Unimolecular decompositions

It has been well established that the six-centered eliminations producing ethylene (C_2H_4) and corresponding acids (see Scheme 2), are predominant unimolecular decomposition channels of ethyl esters. Rate expressions recommended in literature [44] were assigned to these reactions without considering the pressure effects in previous models [6,13]. According to a shock tube study by Ren et al. [15], concentrations of pyrolysis products are sensitive to these unimolecular reactions, particularly at the beginning stages. So model predictions might be impacted if the experiments were conducted under conditions where these reactions are in fall-off regions. Therefore in this work, ab initio based RRKM/master equation analyses were carried out to obtain pressure-dependent rate coefficients for unimolecular decompositions of C_3 – C_5 ethyl esters. The lowest energy conformers for fuel molecules suggested by the references [45–47] were selected as reactants with the fast conformational changes neglected to reduce computational difficulties. Equilibrium geometries for all reactants, transition states (TSs) and products on the potential energy surfaces (PESs) of the three ethyl esters were optimized using the M06-2X functional [48] with the cc-pVTZ basis set [49]. Zero-point energies (ZPE) and vibrational frequencies were scaled by a factor of 0.955 [50]. For more accurate energetic parameters, CCSD(T) energies were extrapolated to complete basis set (CBS) limits using the cc-pVTZ and cc-pVQZ basis sets [51] for stationary points on EF and EA PESs. While for the case of EP, the CCSD(T)/cc-pVQZ calculations were computationally prohibitive. Hence, the CBS limit energies were estimated from extrapolations of the cc-pVDZ and cc-pVTZ results employing the MP2 method [52] to correct the difference in cc-pVDZ, cc-pVTZ and cc-pVQZ based extrapolations [53]. All quantum chemical calculations were performed with the Gaussian 09 program [54]. The PES for EA unimolecular decompositions is demonstrated in Fig. 2 while those for EF and EP are respectively provided in Figs. S1 and S2 in the Supplemental Material.

The phenomenological rate coefficients for all unimolecular decomposition channels were acquired through the master equation analyses with the kinetic code MESS [55] over the temperature range of 500–2000 K and the pressure range of 0.01–100 atm. A conventional transition state theory treatment was employed for the molecular channels whereas the transition state partition functions are evaluated using phase space theory for the bond fission reactions. The torsional modes, including internal rotations of methyl, ethyl and hydroxyl groups, were treated as 1-dimensional (1-D) hindered rotors, with the hindrance potentials

Table 2
The added or updated reactions in the primary mechanisms for C₃–C₅ ethyl esters pyrolysis and combustion in this work. Rate coefficients are given as $k = AT^n \exp(-E_a/RT)$. Units are s⁻¹, cm³ and cal/mol.

No.	Reactions	Pressure	A	n	E _a	Ref.
R1	C ₂ H ₅ OCHO = C ₂ H ₄ + HOCHO	0.01 atm	1.54×10^{49}	-11.15	6.47×10^4	This work ^a
		0.1 atm	6.21×10^{39}	-8.16	6.17×10^4	
		1 atm	2.24×10^{28}	-4.63	5.73×10^4	
		10 atm	7.62×10^{17}	-1.46	5.28×10^4	
		100 atm	1.19×10^{11}	0.59	4.98×10^4	
R2	C ₂ H ₅ OCHO = C ₂ H ₅ OH + CO	0.01 atm	1.25×10^{59}	-14.41	8.22×10^4	This work ^a
		0.1 atm	1.73×10^{55}	-12.74	8.35×10^4	
		1 atm	7.22×10^{43}	-9.01	8.05×10^4	
		10 atm	3.65×10^{29}	-4.59	7.51×10^4	
		100 atm	2.29×10^{18}	-1.19	7.03×10^4	
R3	C ₂ H ₅ OCHO = CH ₃ + CH ₂ OCHO	0.01 atm	5.70×10^{50}	-12.39	9.97×10^4	This work ^a
		0.1 atm	1.45×10^{63}	-15.13	1.04×10^5	
		1 atm	3.29×10^{63}	-14.46	1.07×10^5	
		10 atm	6.02×10^{52}	-10.81	1.05×10^5	
		100 atm	5.92×10^{38}	-6.45	9.96×10^4	
R4	C ₂ H ₅ OCHO = C ₂ H ₅ + OCHO	0.01 atm	6.12×10^{49}	-12.10	1.03×10^5	This work ^a
		0.1 atm	6.67×10^{62}	-15.09	1.06×10^5	
		1 atm	3.34×10^{64}	-14.73	1.09×10^5	
		10 atm	2.63×10^{54}	-11.23	1.07×10^5	
		100 atm	1.53×10^{40}	-6.79	1.02×10^5	
R5	C ₂ H ₅ OCHO + H = PC ₂ H ₄ OCHO + H ₂		5.74×10^{10}	1.20	9.97×10^3	[29]
R6	C ₂ H ₅ OCHO + H = SC ₂ H ₄ OCHO + H ₂		2.16×10^{10}	1.20	6.40×10^3	[29]
R7	C ₂ H ₅ OCHO + H = C ₂ H ₅ OCO + H ₂		2.90×10^{11}	0.96	7.66×10^3	[29]
R8	C ₂ H ₅ OCHO + OH = PC ₂ H ₄ OCHO + H ₂ O		5.84×10^8	1.42	4.37×10^3	This work ^b
R9	C ₂ H ₅ OCHO + OH = SC ₂ H ₄ OCHO + H ₂ O		6.42×10^5	2.06	2.75×10^3	This work ^b
R10	C ₂ H ₅ OCHO + OH = C ₂ H ₅ OCO + H ₂ O		1.29×10^6	2.06	2.75×10^3	[30] ^c
R11	C ₂ H ₅ OCHO + CH ₃ = PC ₂ H ₄ OCHO + CH ₄		8.15×10^7	1.56	1.42×10^4	This work ^b
R12	C ₂ H ₅ OCHO + CH ₃ = SC ₂ H ₄ OCHO + CH ₄		2.52×10^7	1.56	1.21×10^4	This work ^b
R13	C ₂ H ₅ OCHO + CH ₃ = C ₂ H ₅ OCO + CH ₄		9.54×10^{-5}	5.02	8.03×10^3	[30] ^c
R14	C ₂ H ₅ OCHO + HO ₂ = PC ₂ H ₄ OCHO + H ₂ O ₂		7.08×10^9	1.08	2.23×10^4	This work ^b
R15	C ₂ H ₅ OCHO + HO ₂ = SC ₂ H ₄ OCHO + H ₂ O ₂		2.36×10^9	1.08	1.96×10^4	This work ^b
R16	C ₂ H ₅ OCHO + HO ₂ = C ₂ H ₅ OCO + H ₂ O ₂		4.06×10^9	3.68	1.58×10^4	[30] ^c
R17	C ₂ H ₅ OCHO + O = PC ₂ H ₄ OCHO + OH		5.38×10^{10}	0.87	1.23×10^4	This work ^b
R18	C ₂ H ₅ OCHO + O = SC ₂ H ₄ OCHO + OH		1.47×10^{10}	0.87	9.65×10^3	This work ^b
R19	C ₂ H ₅ OCHO + O = C ₂ H ₅ OCO + OH		2.88×10^2	3.32	7.28×10^3	[30] ^c
R20	PC ₂ H ₄ OCHO = C ₂ H ₄ + OCHO	1 Torr	7.93×10^2	1.51	1.52×10^4	[42] ^d
		1 atm	3.52×10^{-34}	11.84	-1.87×10^4	
R21	SC ₂ H ₄ OCHO = CH ₃ CHO + HCO	1 Torr	8.96×10^{13}	-1.61	2.69×10^4	[42] ^d
		1 atm	8.34×10^{27}	-5.19	3.56×10^4	
R22	C ₂ H ₅ OCO = C ₂ H ₅ O + CO	0.05 atm	1.03×10^3	1.29	2.54×10^4	[43]
		1 atm	4.09×10^5	0.81	2.20×10^4	
		10 atm	9.02×10^{14}	-1.72	2.18×10^4	
		100 atm	2.83×10^{22}	-3.44	2.36×10^4	
R23	C ₂ H ₅ OCO = C ₂ H ₅ + CO ₂	0.05 atm	9.72×10^{12}	-1.31	9.42×10^3	[43]
		1 atm	1.25×10^{16}	-1.83	1.13×10^4	
		10 atm	1.04×10^{18}	-2.10	1.28×10^4	
		100 atm	8.69×10^{17}	-1.81	1.37×10^4	
R24	PC ₂ H ₄ OCHO + H = C ₂ H ₄ + HOCHO		1.00×10^{14}	0.00	0.00	This work ^e
R25	SC ₂ H ₄ OCHO + H = C ₂ H ₄ + HOCHO		1.20×10^{14}	0.00	0.00	This work ^e
R26	PC ₂ H ₄ OCHO + CH ₃ = C ₃ H ₆ + HOCHO		2.00×10^{12}	0.00	0.00	This work ^e
R27	SC ₂ H ₄ OCHO + CH ₃ = C ₃ H ₆ + HOCHO		2.00×10^{12}	0.00	0.00	This work ^e
R28	EA = C ₂ H ₄ + CH ₃ COOH	0.01 atm	9.86×10^{39}	-8.39	5.89×10^4	This work ^a
		0.1 atm	2.53×10^{32}	-5.96	5.66×10^4	
		1 atm	3.17×10^{26}	-4.09	5.48×10^4	
		10 atm	7.50×10^{19}	-2.04	5.23×10^4	
		100 atm	1.36×10^{14}	-0.30	4.99×10^4	
R29	EA = CH ₃ + CH ₃ COOCH ₂	0.01 atm	8.06×10^{68}	-17.09	1.04×10^5	This work ^a
		0.1 atm	1.95×10^{69}	-16.39	1.07×10^5	
		1 atm	1.67×10^{62}	-13.73	1.07×10^5	
		10 atm	7.11×10^{51}	-10.35	1.04×10^5	
		100 atm	4.18×10^{40}	-6.87	1.00×10^5	
R30	EA = C ₂ H ₅ + CH ₃ CO ₂	0.01 atm	1.76×10^{69}	-17.13	1.06×10^5	This work ^a
		0.1 atm	6.29×10^{70}	-16.75	1.09×10^5	
		1 atm	1.60×10^{64}	-14.21	1.09×10^5	
		10 atm	7.22×10^{53}	-10.82	1.07×10^5	
		100 atm	2.44×10^{42}	-7.26	1.02×10^5	
R31	EA = CH ₃ + C ₂ H ₅ OCO	0.01 atm	1.81×10^{67}	-17.03	1.07×10^5	This work ^a
		0.1 atm	4.58×10^{69}	-16.89	1.10×10^5	
		1 atm	3.04×10^{63}	-14.46	1.11×10^5	
		10 atm	1.79×10^{53}	-11.09	1.08×10^5	
		100 atm	5.61×10^{41}	-7.52	1.04×10^5	

(continued on next page)

Table 2 (continued)

No.	Reactions	Pressure	A	n	E _a	Ref.
R32	EA = CH ₂ CO + C ₂ H ₅ OH	0.01 atm	1.48×10^{53}	−12.79	8.48×10^4	This work ^a
		0.1 atm	2.33×10^{45}	−10.03	8.38×10^4	
		1 atm	7.74×10^{36}	−7.23	8.18×10^4	
		10 atm	4.96×10^{29}	−4.94	7.96×10^4	
		100 atm	3.95×10^{20}	−2.16	7.60×10^4	
R33	EA = CCOC(OH)*C	0.01 atm	9.83×10^{50}	−12.38	7.94×10^4	This work ^a
		0.1 atm	3.64×10^{41}	−9.15	7.77×10^4	
		1 atm	9.68×10^{32}	−6.30	7.57×10^4	
		10 atm	1.14×10^{27}	−4.36	7.43×10^4	
		100 atm	8.19×10^{16}	−1.25	7.03×10^4	
R34	CCOC(OH)*C = CH ₂ CO + C ₂ H ₅ OH	0.01 atm	3.29×10^{44}	−9.95	5.21×10^4	This work ^a
		0.1 atm	1.37×10^{36}	−7.24	4.97×10^4	
		1 atm	1.27×10^{29}	−5.00	4.75×10^4	
		10 atm	1.89×10^{23}	−3.17	4.56×10^4	
		100 atm	2.13×10^{16}	−1.06	4.28×10^4	
R35	EA + H = EA1 + H ₂		6.21×10^{10}	1.19	9.89×10^3	[29]
R36	EA + H = EA2 + H ₂		1.42×10^{10}	1.22	6.07×10^3	[29]
R37	EA + H = EAm + H ₂		1.44×10^{10}	1.12	7.58×10^3	[29]
R38	EA + OH = EA1 + H ₂ O		6.81×10^0	3.62	-2.73×10^2	[28]
R39	EA + OH = EA2 + H ₂ O		2.68×10^0	3.66	-4.10×10^3	[28]
R40	EA + OH = EAm + H ₂ O		4.53×10^0	3.68	-1.26×10^3	[28]
R41	EA + CH ₃ = EA1 + CH ₄		2.04×10^9	1.09	1.34×10^4	This work ^b
R42	EA + CH ₃ = EA2 + CH ₄		5.41×10^0	3.80	7.16×10^3	This work ^b
R43	EA + CH ₃ = EAm + CH ₄		6.75×10^{-1}	3.80	7.16×10^3	[31] ^c
R44	EA + HO ₂ = EA1 + H ₂ O ₂		4.11×10^{-1}	3.98	9.42×10^3	This work ^b
R45	EA + HO ₂ = EA2 + H ₂ O ₂		3.78×10^{-5}	4.90	5.38×10^3	This work ^b
R46	EA + HO ₂ = EAm + H ₂ O ₂		4.44×10^{-3}	4.41	7.92×10^3	[31] ^c
R47	EA + O = EA1 + OH		2.13×10^5	2.88	1.30×10^4	This work ^b
R48	EA + O = EA2 + OH		1.66×10^5	2.86	9.37×10^3	This work ^b
R49	EA + O = EAm + OH		2.06×10^4	2.86	9.37×10^3	[31] ^c
R50	EA1 = C ₂ H ₄ + CH ₃ CO ₂	1 Torr	7.93×10^2	1.51	1.52×10^4	[42] ^d
		1 atm	3.52×10^{-34}	11.84	-1.87×10^4	
R51	EA2 = CH ₃ CHO + CH ₃ CO	1 Torr	8.96×10^{13}	−1.61	2.69×10^4	[42] ^d
		1 atm	8.34×10^{27}	−5.19	3.56×10^4	
R52	EAm = CH ₂ CO + C ₂ H ₅ O	0.01 atm	5.19×10^{14}	−2.10	4.44×10^4	[34]
		0.04 atm	3.42×10^{22}	−4.13	4.74×10^4	
		1 atm	1.30×10^{38}	−7.91	5.57×10^4	
		10 atm	3.86×10^{39}	−7.84	6.25×10^4	
		100 atm	7.64×10^{32}	−5.56	6.39×10^4	
R53	EA1 + H = C ₂ H ₄ + CH ₃ COOH		1.00×10^{14}	0.00	0.00	This work ^e
R54	EA2 + H = C ₂ H ₄ + CH ₃ COOH		1.20×10^{14}	0.00	0.00	This work ^e
R55	EAm + H = C ₂ H ₄ + CH ₃ COOH		8.00×10^{13}	0.00	0.00	This work ^e
R56	EA1 + CH ₃ = C ₃ H ₆ + CH ₃ COOH		2.00×10^{12}	0.00	0.00	This work ^e
R57	EA2 + CH ₃ = C ₃ H ₆ + CH ₃ COOH		2.00×10^{12}	0.00	0.00	This work ^e
R58	EAm + CH ₃ = C ₂ H ₄ + C ₂ H ₅ COOH		1.00×10^{12}	0.00	0.00	This work ^e
R59	EP = C ₂ H ₄ + C ₂ H ₅ COOH	0.01 atm	4.62×10^{44}	−9.62	6.36×10^4	This work ^a
		0.1 atm	2.92×10^{33}	−6.15	5.93×10^4	
		1 atm	4.33×10^{21}	−2.57	5.43×10^4	
		10 atm	1.77×10^{13}	−0.05	5.05×10^4	
		100 atm	6.93×10^8	1.27	4.85×10^4	
R60	EP = CH ₃ + EAm	0.01 atm	3.16×10^{81}	−20.24	1.09×10^5	This work ^a
		0.1 atm	9.23×10^{69}	−16.31	1.07×10^5	
		1 atm	1.68×10^{52}	−10.76	1.01×10^5	
		10 atm	8.88×10^{35}	−5.81	9.39×10^4	
		100 atm	2.79×10^{25}	−2.65	8.92×10^4	
R61	EP = CH ₃ + CH ₂ OCOC ₂ H ₅	0.01 atm	1.51×10^{83}	−20.74	1.13×10^5	This work ^a
		0.1 atm	1.16×10^{73}	−17.16	1.12×10^5	
		1 atm	2.05×10^{55}	−11.57	1.06×10^5	
		10 atm	1.28×10^{38}	−6.32	9.84×10^4	
		100 atm	4.03×10^{26}	−2.87	9.33×10^4	
R62	EP = C ₂ H ₅ + C ₂ H ₅ CO ₂	0.01 atm	4.00×10^{83}	−20.75	1.12×10^5	This work ^a
		0.1 atm	1.36×10^{73}	−17.08	1.11×10^5	
		1 atm	2.16×10^{55}	−11.48	1.05×10^5	
		10 atm	1.85×10^{38}	−6.28	9.77×10^4	
		100 atm	8.33×10^{26}	−2.87	9.26×10^4	
R63	EP = C ₂ H ₅ + C ₂ H ₅ OCO	0.01 atm	3.76×10^{84}	−21.20	1.15×10^5	This work ^a
		0.1 atm	2.42×10^{75}	−17.86	1.15×10^5	
		1 atm	6.16×10^{57}	−12.30	1.09×10^5	
		10 atm	1.55×10^{40}	−6.92	1.01×10^5	
		100 atm	1.62×10^{28}	−3.32	9.61×10^4	
R64	EP = C ₂ H ₅ O + C ₂ H ₅ CO	0.01 atm	6.12×10^{84}	−21.66	1.21×10^5	This work ^a
		0.1 atm	1.93×10^{79}	−19.29	1.22×10^5	
		1 atm	3.96×10^{62}	−13.90	1.17×10^5	
		10 atm	4.13×10^{43}	−8.08	1.10×10^5	
		100 atm	5.51×10^{29}	−3.89	1.04×10^5	

(continued on next page)

Table 2 (continued)

No.	Reactions	Pressure	A	n	E _a	Ref.
R65	EP = CH ₃ CHCO + C ₂ H ₅ OH	0.01 atm	3.08 × 10 ⁶⁵	−16.03	9.49 × 10 ⁴	This work ^a
		0.1 atm	8.90 × 10 ⁵²	−11.97	9.14 × 10 ⁴	
		1 atm	1.66 × 10 ³⁶	−6.82	8.45 × 10 ⁴	
		10 atm	6.20 × 10 ²¹	−2.46	7.86 × 10 ⁴	
		100 atm	1.18 × 10 ¹³	0.15	7.47 × 10 ⁴	
R66	EP = CCOC(OH)*CC	0.01 atm	3.35 × 10 ⁶¹	−15.22	8.75 × 10 ⁴	This work ^a
		0.1 atm	7.17 × 10 ⁴⁹	−11.43	8.44 × 10 ⁴	
		1 atm	2.46 × 10 ³⁴	−6.64	7.86 × 10 ⁴	
		10 atm	1.38 × 10 ¹⁹	−2.02	7.20 × 10 ⁴	
		100 atm	3.57 × 10 ⁹	0.86	6.77 × 10 ⁴	
R67	CCOC(OH)*CC = CH ₃ CHCO + C ₂ H ₅ OH	0.01 atm	6.30 × 10 ⁴⁴	−10.00	5.26 × 10 ⁴	This work ^a
		0.1 atm	3.08 × 10 ³⁵	−7.05	4.96 × 10 ⁴	
		1 atm	7.43 × 10 ²⁴	−3.78	4.54 × 10 ⁴	
		10 atm	1.75 × 10 ¹⁵	−0.88	4.12 × 10 ⁴	
		100 atm	2.18 × 10 ¹⁰	0.59	3.89 × 10 ⁴	
R68	EP + H = EPMJ + H ₂		6.45 × 10 ¹⁰	1.21	9.87 × 10 ³	[29]
R69	EP + H = EPEJ + H ₂		1.73 × 10 ¹⁰	1.24	5.99 × 10 ³	[29]
R70	EP + H = EP3J + H ₂		1.28 × 10 ¹¹	1.19	9.32 × 10 ³	[29]
R71	EP + H = EP2J + H ₂		1.69 × 10 ¹⁰	1.21	4.87 × 10 ³	[29]
R72	EP + OH = EPMJ + H ₂ O		4.19 × 10 ¹	3.72	−1.90 × 10 ¹	This work ^b
R73	EP + OH = EPEJ + H ₂ O		4.63 × 10 ^{−2}	4.32	−1.64 × 10 ³	This work ^b
R74	EP + OH = EP3J + H ₂ O		1.91 × 10 ¹	3.66	−6.58 × 10 ²	[32] ^c
R75	EP + OH = EP2J + H ₂ O		9.15 × 10 ¹	3.23	−1.54 × 10 ³	[32] ^c
R76	EP + CH ₃ = EPMJ + CH ₄		1.77 × 10 ^{−16}	8.57	2.64 × 10 ³	This work ^b
R77	EP + CH ₃ = EPEJ + CH ₄		4.70 × 10 ^{−25}	11.28	−3.62 × 10 ³	This work ^b
R78	EP + CH ₃ = EP3J + CH ₄		8.79 × 10 ^{−4}	4.68	8.37 × 10 ⁴	[32] ^c
R79	EP + CH ₃ = EP2J + CH ₄		7.82 × 10 ^{−5}	4.92	3.38 × 10 ³	[32] ^c
R80	EP + HO ₂ = EPMJ + H ₂ O ₂		1.95 × 10 ^{−16}	8.42	1.16 × 10 ⁴	This work ^b
R81	EP + HO ₂ = EPEJ + H ₂ O ₂		5.92 × 10 ^{−17}	8.42	5.61 × 10 ³	This work ^b
R82	EP + HO ₂ = EP3J + H ₂ O ₂		2.71 × 10 ^{−7}	5.74	1.15 × 10 ⁴	[32] ^c
R83	EP + HO ₂ = EP2J + H ₂ O ₂		3.54 × 10 ^{−8}	5.87	9.48 × 10 ³	[32] ^c
R84	EP + O = EPMJ + OH		1.07 × 10 ³	3.60	6.56 × 10 ³	This work ^b
R85	EP + O = EPEJ + OH		2.91 × 10 ²	3.60	3.89 × 10 ³	This work ^b
R86	EP + O = EP3J + OH		1.78 × 10 ⁰	4.01	3.42 × 10 ³	[32] ^c
R87	EP + O = EP2J + OH		4.55 × 10 ¹	3.60	1.97 × 10 ³	[32] ^c
R88	CH ₂ OCOC ₂ H ₅ = C ₂ H ₅ CO + CH ₂ O	0.01 atm	6.29 × 10 ³⁵	−7.51	3.73 × 10 ⁴	[10]
		0.04 atm	2.49 × 10 ³⁷	−7.79	3.90 × 10 ⁴	
		1 atm	4.09 × 10 ³⁶	−7.17	4.12 × 10 ⁴	
		10 atm	3.93 × 10 ³⁰	−5.19	3.97 × 10 ⁴	
		100 atm	1.09 × 10 ²²	−2.56	3.63 × 10 ⁴	
R89	EPMJ = C ₂ H ₄ + C ₂ H ₅ CO ₂	1 Torr	7.93 × 10 ²	1.51	1.52 × 10 ⁴	[42] ^d
		1 atm	3.52 × 10 ^{−34}	11.84	−1.87 × 10 ⁴	
R90	EAEJ = CH ₃ CHO + C ₂ H ₅ CO	1 Torr	8.96 × 10 ¹³	−1.61	2.69 × 10 ⁴	[42] ^d
		1 atm	8.34 × 10 ²⁷	−5.19	3.56 × 10 ⁴	
R91	EP3J = C ₂ H ₄ + C ₂ H ₅ OCO	0.01 atm	1.66 × 10 ³⁶	−7.70	3.51 × 10 ⁴	[10]
		0.04 atm	7.33 × 10 ³⁸	−8.27	3.73 × 10 ⁴	
		1 atm	4.32 × 10 ⁴¹	−8.61	4.15 × 10 ⁴	
		10 atm	1.16 × 10 ³⁸	−7.28	4.16 × 10 ⁴	
		100 atm	2.21 × 10 ³⁰	−4.82	3.93 × 10 ⁴	
R92	EP2J = CH ₃ CHCO + C ₂ H ₅ O	0.01 atm	7.31 × 10 ⁴⁴	−10.59	5.93 × 10 ⁴	[10]
		0.04 atm	8.48 × 10 ⁵⁰	−12.02	6.35 × 10 ⁴	
		1 atm	4.92 × 10 ⁵¹	−11.52	6.80 × 10 ⁴	
		10 atm	4.64 × 10 ⁴²	−8.52	6.60 × 10 ⁴	
		100 atm	1.87 × 10 ³⁰	−4.69	6.12 × 10 ⁴	
R93	EPMJ + H = C ₂ H ₄ + C ₂ H ₅ COOH		5.00 × 10 ¹³	0.00	0.00	This work ^e
R94	EPEJ + H = C ₂ H ₄ + C ₂ H ₅ COOH		5.00 × 10 ¹³	0.00	0.00	This work ^e
R95	EP3J + H = C ₂ H ₄ + C ₂ H ₅ COOH		5.00 × 10 ¹³	0.00	0.00	This work ^e
R96	EP2J + H = C ₂ H ₄ + C ₂ H ₅ COOH		5.00 × 10 ¹³	0.00	0.00	This work ^e
R97	EPMJ + CH ₃ = C ₃ H ₆ + C ₂ H ₅ COOH		1.00 × 10 ¹²	0.00	0.00	This work ^e
R98	EPEJ + CH ₃ = C ₃ H ₆ + C ₂ H ₅ COOH		1.00 × 10 ¹²	0.00	0.00	This work ^e

Notes: C₂H₅OCHO, CH₃CH₂OC(=O)H; PC₂H₄OCHO, CH₂CH₂OC(=O)H; PC₂H₄OCHO, CH₃CHOC(=O)H; C₂H₅OCO, C₂H₅OC(=O); HOCHO, HC(=O)OH; EA, CH₃CH₂OC(=O)CH₃; CCOC(OH)*C, CH₃CH₂OC(OH)=CH₂; EA1, CH₂CH₂OC(=O)CH₃; EA1, CH₂CH₂OC(=O)CH₃; EA2, CH₃CHOC(=O)CH₃; EAm, CH₃CH₂OC(=O)CH₂; CH₃CO₂, CH₃C(=O)O; EP, CH₃CH₂OC(=O)CH₂CH₃; CCOC(OH)*CC, CH₃CH₂OC(OH)=CHCH₃; EPMJ, CH₂CH₂OC(=O)CH₂CH₃; EPEJ, CH₃CH₂OC(=O)CH₂CH₃; EP3J, CH₃CH₂OC(=O)CH₂CH₂; EP2J, CH₃CH₂OC(=O)CHCH₃; C₂H₅CO₂, CH₃CH₂C(=O)O.

^a Theoretical calculations.

^b Ratestrategy, see text in Section 3.2.

^c Analogies to methyl ester reactions.

^d Analogies to β-scissions of ethanol radicals.

^e Estimations.

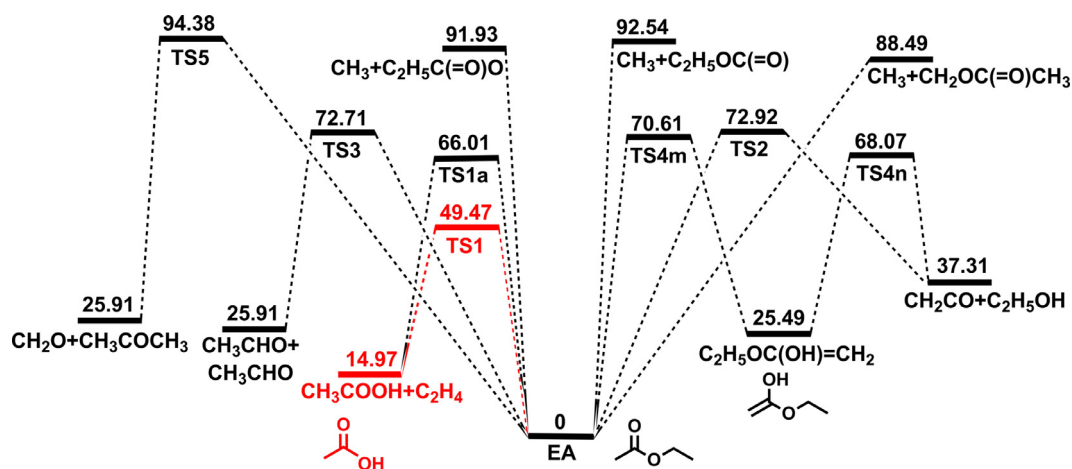


Fig. 2. Calculated potential energy surface (PES) for ethyl acetate (EA) unimolecular decompositions at the level of CCSD(T)/CBS//M06-2X/cc-pVTZ (Unit, kcal/mol).

relaxedly scanned at the B3LYP/6-311G(d) level. The asymmetric Eckart correction [56] was incorporated to compensate for the tunneling effects. Interactions between reactants and the bath argon were modeled with the Lennard–Jones (L–J) potential. The L–J parameters for the three ethyl esters were correlated with the critical temperatures and pressures (EF: $T_c = 508.4$ K, $p_c = 47.4$ bar; EA: $T_c = 523.3$ K, $p_c = 38.8$ bar; EP: $T_c = 546.0$ K, $p_c = 33.6$ bar [57]). Single-parameter temperature dependent exponential down models, $\langle \Delta E \rangle_{\text{down}} = \alpha_0 \times (T/298)^n \text{ cm}^{-1}$, were used to represent the collisional energy transfer functions. The parameters of α_0 and n for EA ($\text{C}_4\text{H}_8\text{O}_2$) were respectively set as 200 cm^{-1} and 0.85 which were adopted for the unimolecular system of methyl propionate (an isomer of EA) [58]. And according to a recent theoretical work by Jasper et al. [59], the amount of energy transferred per time through collision generally increases with the number of heavy atoms, so we assigned α_0 of 150 cm^{-1} to EF ($\text{C}_3\text{H}_6\text{O}_2$) and 250 cm^{-1} to EP ($\text{C}_5\text{H}_{10}\text{O}_2$) based on their molecular weights. And further analyses indicated that for all the three ethyl esters, a change of $\pm 50 \text{ cm}^{-1}$ in α_0 will bring a difference within 30% to the rate coefficients of the major elimination pathways, which has already been considered in uncertainty estimations for calculated results.

The three PESs (see Figs. 2, S1 and S2) share a similarity that the channels leading to C_2H_4 and acids, as highlighted in separate PESs, are of energy barriers around 50 kcal/mol , at least 15 kcal/mol lower than those of other competing routes. Therefore the dominant roles of these elimination reactions in ethyl ester decompositions can be expected. Actually, the current RRKM/master equation analyses show that they take branching ratios of over 90% under all investigated conditions. Pressure-dependent rate coefficients for the three leading elimination reactions are demonstrated in Fig. 3, and the values adopted in Westbrook models [6,13] and available measurements [60–62] and calculations [63,64] in literature are also shown for comparison. Most experimental data points were obtained at low-to-medium temperatures below 1000 K and only the high pressure limit (HPL) rate constants were calculated in previous theoretical works [63,64]. For the most studied EA reaction, the majority of measurements collected by the NIST database [61] are in good consistence though they were obtained by different methods under various pressures. It indicates that rate coefficients for this reaction are of small uncertainties and barely affected by the fall-off effects at low-to-medium temperatures. According to Fig. 3, for all the three reactions, our calculations can well match the available measurements and at elevated temperatures over 1000 K the current calculated HPL rates are very close to the values adopted in Westbrook models [6,13]. The current calculated results indicate that fall-off effects cannot be neglected at

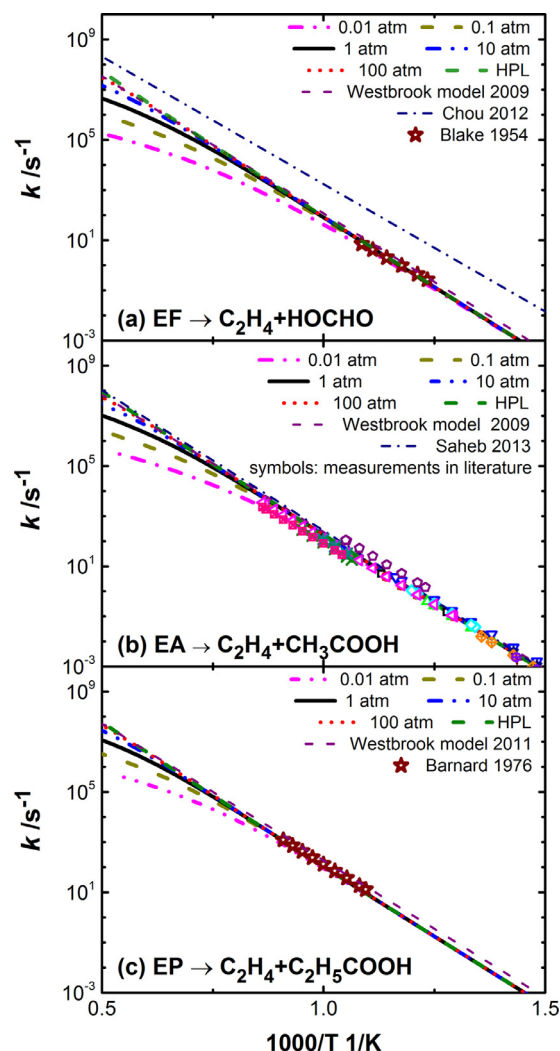


Fig. 3. Comparisons of the rate coefficients for the energy-favored six-membered eliminations of (a) ethyl formate (EF), (b) ethyl acetate (EA) and (c) ethyl propionate (EP). Symbols: measurements in literature ([60] for EF, [61] for EA and [62] for EP); Thick lines: calculations in the present work, thin dash lines: the values adopted in the models of Westbrook et al. [6,13], thin dash-dot lines: calculated HPL in literature ([64] for EF and [63] for EA).

conventional combustion temperatures around 2000 K, because a difference of more than an order of magnitude can be found between the rate coefficients at 0.01 atm and the HPL. Therefore the previous models using the HPL rate constants for these crucial eliminations might not appropriately interpret fuel consumption patterns under low-pressure premixed flame conditions.

The dominant eliminations as well as channels of minor contributions ($\geq 2\%$) under any conditions of current calculations were integrated into the model, with the calculated rate coefficients fitted in the PLOG format, as R1–R4, R28–R34 and R59–R67 listed in Table 2.

3.2. Hydrogen abstractions

In the modeling works by Westbrook et al. [6,13], a basic rule was followed that hydrogen abstractions from similar sites by each radical reactant have identical rates [6]. The reasonability of such rate rules have been supported by several recent theoretical works [27–29]. Therefore calculated rate coefficients for hydrogen abstractions from the aliphatic chains of methyl esters can be assigned to corresponding reactions of ethyl esters, given that several systematical high-level calculations for C_2 – C_4 methyl esters [30–32] have been available in literature. However, there exists a risk that model predictions could deteriorate if only rate coefficients for hydrogen abstractions from “one side” (the acid moiety) of the ethyl ester molecules are updated, because the branching ratios between hydrogen abstractions from different sites may change much due to the inconsistent origins of kinetic parameters. The importance of considering the relative reaction rates in optimizing detailed kinetic models has been addressed in a recent work [65]. So in the present work, priorities were given to employing systematical calculated rate coefficients for hydrogen abstractions of ethyl esters, including those from EA by HO_2 [27] and OH [28], and from the three esters by H [29]. For the rest reactions of this category, a balanced strategy was taken that the rates of hydrogen abstractions from α' (or formyl for EF) and β' sites (see Scheme 1) were set to be equal with those of corresponding methyl ester reactions [30–32], while for hydrogen abstractions from α and β sites (see Scheme 1), the rate coefficients were respectively scaled to maintain their original branching ratios with those from α' sites in Westbrook models [6,13]. Systematical calculations are still advocated for accurate and consistent kinetic parameters of hydrogen abstractions from ethyl esters.

3.3. Radical consumptions

In previous models [6,13,15], the fates of fuel radicals yielded from hydrogen abstractions were always interpreted as exclusively dissociating through the energy-favored β -scissions. And only HPLs were assigned to these reactions because they were considered as rapid and non-rate-limiting steps. A recent theoretical study [33] has proposed that under low-pressure flame conditions, β -scissions of relatively stable fuel radicals can be inhibited by the fall-off effects. These fuel radicals can therefore accumulate and go through well-skipping (addition-elimination) reactions together with flame radicals like H and CH_3 [33]. After adding these reactions to kinetic models for ethanol and methyl formate, pronounced improvements were observed in predictions for mole fractions of specific species in low-pressure flames [33]. So fuel radical + H and + CH_3 well-skipping reactions (R24–R27, R53–R58 and R93–R98) are incorporated in the current model for simulations of premixed flame structures. And according to theoretical case studies by Labbe et al. [33], rate constants for the dominant channels for fuel radical + H/ CH_3 well-skipping reactions respectively cover scale ranges of 10^{13} – 10^{14} and 10^{11} – 10^{13} mol/cm³/s in low-pressure (<0.1 atm) combustion environments.

Rate coefficients for fuel radical β -scissions were accordingly updated with pressure-dependent values. For decompositions of α' and β' radicals produced as hydrogen atoms in alkanolic acid chains are abstracted, rate coefficients were assigned with those for corresponding methyl ester reactions [10,34]. And for the cases of α and β radicals formed from hydrogen abstractions from ethoxy moieties, analogies were made to the β -scissions of ethanol radicals [42]. An exception was made for the formyl radical of EF [$C_2H_5O\dot{C}(=O)$], a short-lived radical similar to the $CH_3O\dot{C}(=O)$ radical which is consumed through two competing channels respectively leading to $CH_3 + CO_2$ and $CH_3O + CO$ [5,33,43]. The former takes an overwhelming branching ratio of over 99% according to theoretical calculations by Klippenstein [43]. Calculated rate coefficients for $CH_3O\dot{C}(=O)$ decompositions [43] were applied to the similar reaction pair of $C_2H_5O\dot{C}(=O) \rightarrow C_2H_5 + CO_2$ and $C_2H_5O\dot{C}(=O) \rightarrow C_2H_5O + CO$ in the current model.

3.4. Sub-mechanisms for acids

Acids are crucial intermediates produced from both unimolecular eliminations and fuel radical + H/ CH_3 addition-elimination reactions as mentioned above, whose consumptions perpetuate reaction sequences in ethyl ester systems. So the sub-mechanisms for C_1 – C_3 acids were also updated based on experimental and theoretical efforts in literature as well as in the current work.

A detailed kinetic model for formic acid ($HOCHO$) high-temperature oxidation was recently developed by Marshall and Glarborg [35], in which ab initio calculations were performed to derive rate coefficients for key hydrogen abstractions and thermodynamic data for $HOCHO$ and relevant radicals were revised. This $HOCHO$ sub-mechanism tested with available flame speed measurements [35] was adopted in the present model, with two modifications made. One is for the competing dehydration ($H_2O + CO$) and decarboxylation ($CO_2 + OH$) pathways in the unimolecular decomposition of $HOCHO$. In a recent shock tube study [36], rates for the two channels were directly measured at elevated temperatures of 1230–1821 K and two different pressures of 1 and 6.5 atm. Fitted rate coefficients by Elwardany et al. [36] were adopted in the current model, whose uncertainties were claimed to be within 50% [36]. The other revision was for the consumption of $HO\dot{C}(=O)$ radical which goes through two possible pathways respectively leading to $H + CO_2$ and $OH + CO$. Such a reaction scheme is quite similar to those for the above-mentioned $CH_3O\dot{C}(=O)$ and $C_2H_5O\dot{C}(=O)$ radicals, so the same calculated rate coefficients [43] were applied to the branching decompositions of $HO\dot{C}(=O)$ radical.

Christensen and Konnov [37] recently proposed a detailed reaction mechanism for acetic acid (CH_3COOH) based on an extensive literature review, providing uncertainty factors for rate coefficients of all CH_3COOH related reactions. We integrated this CH_3COOH sub-mechanism [37] to our model after revising the unimolecular decomposition part. In the original sub-mechanism [37], several intra-molecular elimination pathways were considered for CH_3COOH , and the rate coefficients at the HPL were originated from a theoretical work by Clark et al. [66], which highlighted the competition between dehydration and decarboxylation channels for carboxylic acids decompositions. Bond fissions were neglected in the calculations [66] and thus excluded from the model by Christensen and Konnov [37]. However, at high temperatures bond fission reactions may lead to the formation of radicals playing as important reactivity improvers. Thus in this work the unimolecular decomposition of CH_3COOH , including both intra-molecular eliminations and bond fissions, were explored with the same theoretical methods mentioned in Section 3.1 and the resulted PESs are provided in Fig. S3 in the Supplemental Material. The calculated pressure-dependent rate coefficients for the dehydration ($CH_3COOH \rightarrow H_2O + CH_2CO$) and the decarboxyla-

tion ($\text{CH}_3\text{COOH} \rightarrow \text{CO}_2 + \text{CH}_4$) were compared with the calculated results in [66] and measurements in literature [36,67–69], as exhibited in Fig. S4. The HPL rate constants calculated in the current work agree well with those by Clark et al. [66]. For the dehydration reaction, calculated rate coefficients are lower than measurements because experimental results were derived from the concentrations of H_2O which is actually mainly produced through a two-step process, $\text{CH}_3\text{COOH} \rightarrow \text{CH}_2=\text{C}(\text{OH})_2 \rightarrow \text{H}_2\text{O} + \text{CH}_2\text{CO}$, instead of the direct elimination. This has been suggested by theoretical efforts in [66] as well as the current work, but cannot be revealed by previous experimental works. While for the decarboxylation, rate constants from different origins show a good consistency.

Very limited studies are available in literature revealing the reaction kinetics of propionic acid ($\text{C}_2\text{H}_5\text{COOH}$). Mendes et al. [38] reported calculated rate coefficients for hydrogen abstractions from $\text{C}_2\text{H}_5\text{COOH}$ by H , OH , HO_2 and CH_3 , which were employed in the current model. Theoretical calculations were performed to obtain rate coefficients for $\text{C}_2\text{H}_5\text{COOH}$ unimolecular decompositions with details provided in the Supplemental Material. Compared to the cases of HOCHO and CH_3COOH , bond fissions play a more important role in the decomposition of $\text{C}_2\text{H}_5\text{COOH}$ as $\text{C}_2\text{H}_5\text{COOH} \rightarrow \text{CH}_3 + \text{CH}_2\text{C}(=\text{O})\text{OH}$ is found to be the dominant route at high temperatures. It is noticed that methyl ketene (CH_3CHCO) can be easily formed from consumptions of both $\text{C}_2\text{H}_5\text{COOH}$ and EP , but thermal decomposition pathways are missing from the consumption mechanism of this relatively unstable species in previous models. So a unimolecular decomposition pathway, $\text{CH}_3\text{CHCO} \rightarrow \text{C}_2\text{H}_4 + \text{CO}$, was added to the current model, with the rate coefficient assumed to be the same with that of a similar reaction for ketene ($\text{CH}_2\text{CO} \rightarrow \text{CH}_2 + \text{CO}$) [70], following the suggestion by Lam et al. [71].

In total, 281 species and 1736 reactions are included in the current model which is aimed at interpreting high-temperature combustion kinetics of C_3 – C_5 ethyl esters. Thermochemical quantities were inherited from the Westbrook models [6,13] in which the program THERM [72] was employed for the calculations. The entire model containing kinetic, thermochemical and transport parameters can be found in the Supplemental Material. All simulations in this work were performed with the Chemkin Pro [73] package. The reflected SHOCK application was selected to simulate species concentration time histories during ethyl esters pyrolysis in a shock tube [15], with the temperatures and pressures behind reflected shock waves inputted. While for simulations of premixed flame structures, the PREMIX module was employed. The measured temperature profiles were taken as input, and the mixture-averaged transport and the thermal diffusion (Soret effects) were considered. The grid numbers were increased till converged, grid-independent solutions were obtained with the gradient and curvature tolerances (GRAD and CURV) set as 0.1 and 0.05, respectively.

4. Results and discussion

Speciation information can provide overall validations for the kinetic model as well as effective constraints on certain reactions. In this section, the current model will be tested with the flame structure measurements accomplished in this work and the species concentration time-histories during pyrolysis reported by Ren et al. [15]. The roles of some key reactions in fuel consumptions and intermediate formations will be discussed.

4.1. Low-pressure premixed flames

To facilitate the discussion in this section, the initial fuel consumption steps are demonstrated first in Fig. 4, based on rate of production (ROP) analyses in entire flame ranges with the validated model. The reaction pathways in Fig. 4 include the consump-

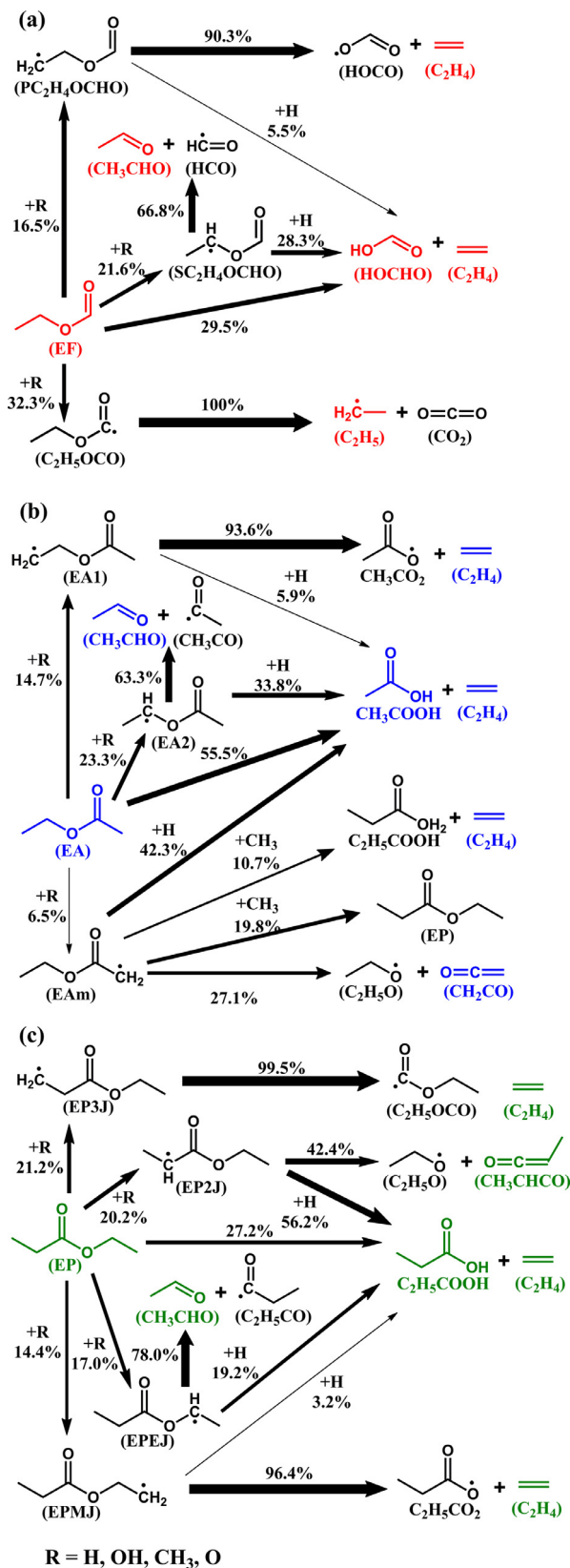


Fig. 4. Initial fuel consumption pathways for the three ethyl esters in premixed flames based on the integrated ROP analyses along the entire flames. The numbers are percent contribution to the consumption of the species on the source side and only reactions with percentages larger than 3% are shown.

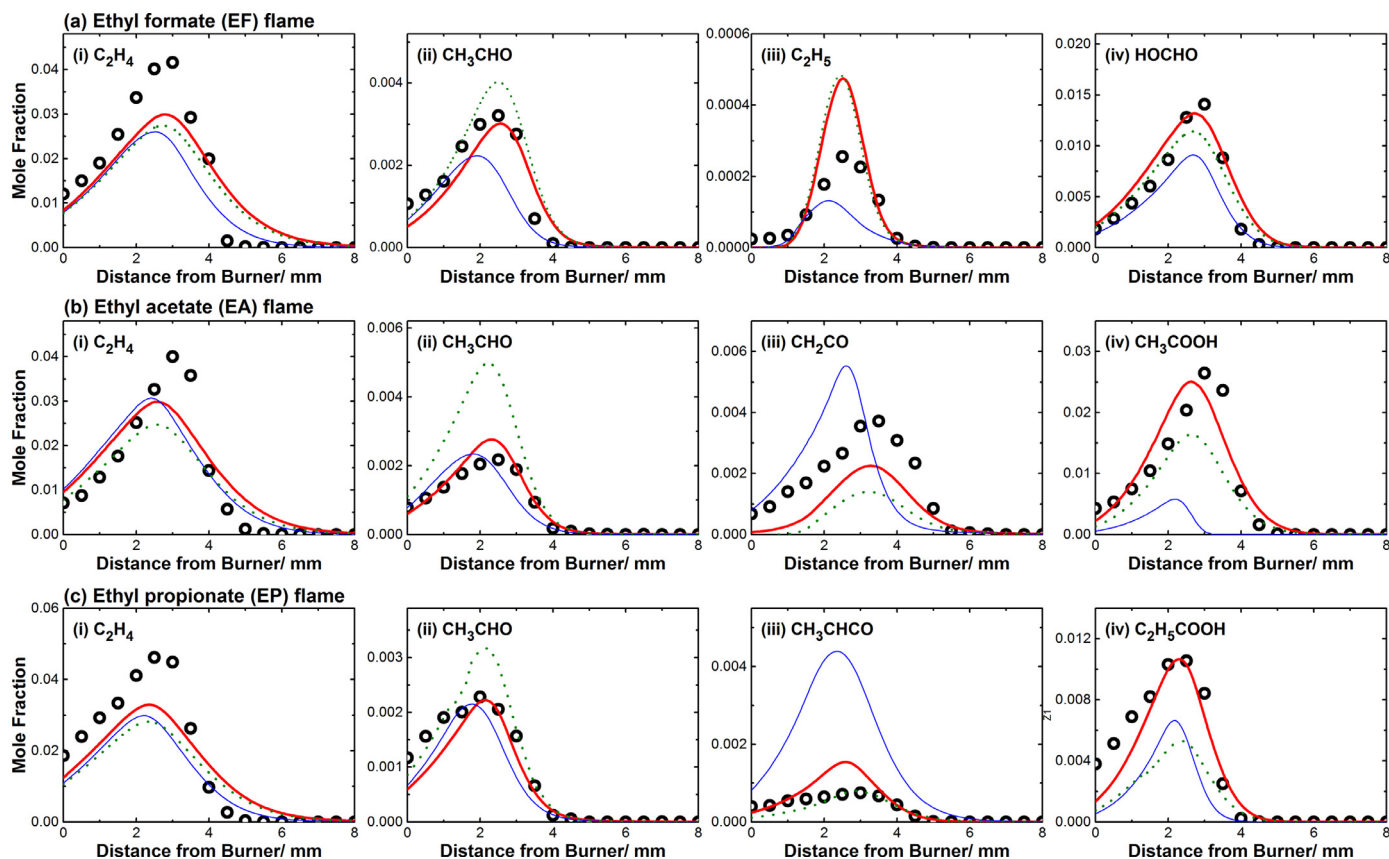


Fig. 5. Mole fraction profiles of measured primary products in premixed flames of the three ethyl esters. Symbols: experimental measurements, thick red solid lines: the present model, thin blue solid lines: Westbrook model [13], green dash lines, the present model excluding well-skipping reactions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

tion channels of fuels and fuel radicals, and the resulting products are defined as “primary species” in this work. And those detected in the current experiments are highlighted in Fig. 4, whose experimental and simulated mole fraction profiles are displayed in Fig. 5. As can be seen, C_2H_4 and CH_3CHO are two primary species shared by all the three flames, because both intermediates are produced following hydrogen abstractions from the common ethyl (CH_3CH_2-) moiety of ethyl esters. In term of the fuel consumption, besides hydrogen abstractions which are considered as the major channels under flame conditions, the six-membered eliminations leading to C_2H_4 and acids are also found significantly responsible, accounting for about 30% of fuel consumptions in the EF and EP flames, while nearly 60% in the case of EA (see Fig. 4). In an early flow reactor study of ethyl butanoate oxidation [74], the elimination producing C_2H_4 and butanoic acid (C_3H_7COOH) was even found to account for 98% of the fuel consumption at 943 K and 1 atm, demonstrating the importance of these intra-molecular eliminations.

To gain insights into the different competitive consequences between eliminations and hydrogen abstractions by H atoms in the three flames, rate coefficients of these two types of reactions were respectively compared among the C_3 – C_5 ethyl esters. The comparisons exhibited in Fig. S5 cover temperatures from 700 to 1800 K, corresponding to those in regions near the burner where fuels are consumed. Given the simulated results that the H atom concentrations are very similar (differences within 20%) in the flame zones, the primary reason for the dominance of the elimination in the EA consumption lies in the smallest overall rate coefficient of hydrogen abstraction by H atom [29] from EA. Because C–C bond in the $[C(=O)-CH_3]$ group is weakened by the adjacent C=O, the hydrogen atoms from the same site become hard to be abstracted. In

addition, according to the currently calculated rate coefficients, the predominant eliminations leading to EA and EP dissociations occur at faster paces than that in the case of EF. Discrepancies in absolute values of the rate coefficients result in different relative competitiveness of the two reaction categories for fuel consumptions in these three flames. The outstanding role of the elimination reaction in the EA flame can also be proven by the higher concentrations of CH_3COOH measured in the EA flame, compared to $HOCHO$ and C_2H_5COOH respectively in EF and EP flames, as can be seen in Fig. 5.

For the primary species illustrated in Fig. 5, besides the predictions by the current model, simulations with the Westbrook model [13] are also shown for comparison. The current model can better capture measurements, which could be attributed to the appropriate kinetic parameters for fuel consumption reactions acquired from theoretical calculations in literature [28–32,38] as well as in the current work. And the inclusion of well-skipping reactions also helps in improving predictions for the primary species. As the thin dash lines shown in Fig. 5, simulations will be degraded if excluding well-skipping reactions from current model. As mentioned above, the currently adopted rate coefficients were estimated based on a relevant theoretical work [33]. Meanwhile they were determined in reasonable ranges to best improve the model predictions. Further theoretical efforts are desired to enhance the accuracy of these kinetic parameters because this reaction category potentially improves model predictions for low-pressure flame structures. It can be noted from the reaction pathways in Fig. 4 that the fates of fuel radicals are actually changed by including well-skipping reactions. Instead of exclusively decomposing, part of fuel radicals combine with H and CH_3 leading

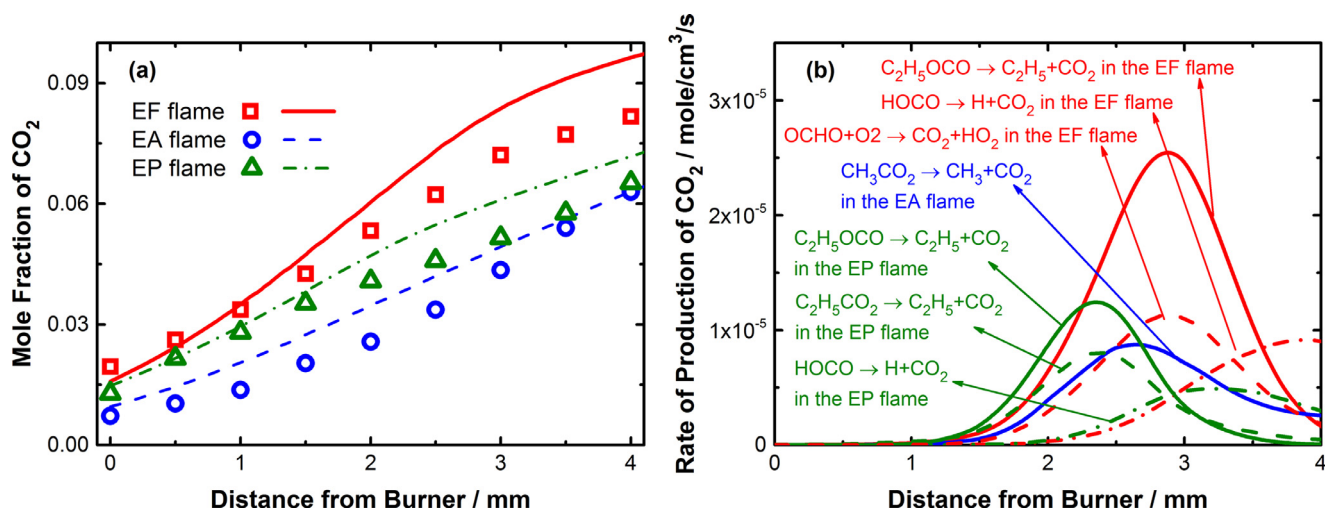


Fig. 6. (a) CO_2 mole fraction profiles and (b) ROP coefficients for dominant CO_2 contributors in the flame zones of the premixed flames of the three ethyl esters. Symbols: measurements; Lines: simulations.

to molecular products. In this way, the reaction flux from fuel radicals is rearranged to different primary species. For example, a portion of the carbon flux to CH_3CHO , a crucial primary species formed from β -scissions of the β fuel radicals [$\text{CH}_3\dot{\text{C}}\text{HOC}(=\text{O})\text{R}$], is shifted to C_2H_4 and acids. An unexpected change is found in the concentrations of CH_2CO in the EA flame, which is identified as a primary species because it can be directly produced from the β -scission of the α' fuel radical [$\text{CH}_3\text{CH}_2\text{CO}(=\text{O})\dot{\text{C}}\text{H}_2$] of EA. When distracting part of the carbon flux to the well-skipping routes, the peak mole fraction of CH_2CO is however increased. ROP analysis was therefore performed for CH_2CO , with the results indicating that the main source (over 70%) of the CH_2CO formation is the decomposition of $\dot{\text{C}}\text{H}_2\text{COOH}$ radical. Therefore the increase in CH_2CO production is originated from the enhanced concentration of CH_3COOH , which is in accord with the dominance of the six-membered elimination in the fuel consumption and the remarkable level of CH_3COOH in the EA flame.

Some primary species are too short-lived or reactive to be detected in the experiments, but they are key intermediates in the formation of major combustion products. For example, several CO_2 precursors can be recognized in Fig. 4, including $\text{C}_2\text{H}_5\text{OCO}$ and HOCO in the EF flame, CH_3CO_2 in the EA flame, and $\text{C}_2\text{H}_5\text{OCO}$ and $\text{C}_2\text{H}_5\text{CO}_2$ in the EP flame. All of them decompose almost exclusively through channels leading to CO_2 formation, as explained in the modeling section. The CO_2 mole fractions in flame regions (0–4 mm) are shown in Fig. 6(a), including both measured and simulated results. And according to ROP analyses for CO_2 in separate flames displayed in Fig. 6(b), decompositions of the above-mentioned radicals are exactly the dominant CO_2 contributors in the flame regions. The highest carbon flux from fuel to CO_2 precursors in the EF flame, as can be seen from Fig. 4, accounts for the highest CO_2 production rate in the EF flame and consequently the fastest CO_2 rising pace in the flame zone. The mole fraction distributions of CO_2 , together with those of other major species (fuels, O_2 , Ar, CO, H_2O , H_2) in entire ranges of separate flames are provided in Fig. S6 in the Supplemental Material. Besides the good predictive capability of the kinetic model for major species concentrations, different CO/ CO_2 ratios in the three flames can be observed, which is in consistent with the fact that specific fuel-related reactions lead to CO and CO_2 formations at different rates in flame zones of the three flames. While in the equilibrium post flame zone, a balance between CO and CO_2 can be struck by the reaction $\text{CO} + \text{OH} = \text{CO}_2 + \text{H}$ and the final CO/ CO_2 ratio is mainly determined by the initial C/O ratio of the flame mixtures.

The model performances were also tested with measurements for other typical combustion species which are involved in the reactions of the secondary and the base C_0 – C_4 mechanisms. Species mole fraction profiles in the EA flame are shown in Fig. 7 while those in EF and EP flames are shown in Figs. S7 and S8. The current model improves the predictions for C_2 hydrocarbons compared to the Westbrook model [13]. Generally, the three flames have similar species pools regarding to the types of conventional combustion intermediates. Hydrocarbon species larger than C_4 were under the detection limit and considerable portions of the fuel carbon atoms end up in oxygenated species. While the amounts of specific intermediates vary among the three flames. For example, it can be noted that CH_4 and C_2H_6 , which peak close to the burner, are of the highest concentrations in the EA flame. This can be explained by the ROP-analyzed results that CH_4 and C_2H_6 are formed predominantly from CH_3 reactions, $\text{CH}_3 + \text{H}(+\text{M}) \rightarrow \text{CH}_4(+\text{M})$ and $\text{CH}_3 + \text{CH}_3(+\text{M}) \rightarrow \text{C}_2\text{H}_6(+\text{M})$, and CH_3 can be easily released from two fuel-related radicals CH_3CO and CH_3CO_2 in the EA flame. CH_3 however peaks relatively downstream with similar peak concentrations in the three flames, because CH_3 is involved in numerous reactions in small radical pools and the concentrations can be influenced by flame conditions such as the stoichiometry and temperatures, which are similar in the three flames.

4.2. Shock tube pyrolysis

The concentration time histories of CO, CO_2 and H_2O during shock tube pyrolysis of the three ethyl esters [15] are important targets for the model validation because these measurements are sensitive to ethyl esters thermal decomposition and acids consumption reactions [15] which can therefore be examined with pertinence. The simulations were performed using the current model as well as the model proposed by Ren et al. [15] based on the Westbrook models [6,13]. According to the experimental and simulated results shown in Fig. 8, both models can give satisfactory predictions for the measurements [15] during EF and EA pyrolysis, while in the case of EP, the current model can better reproduce the measured concentration variations of the three pyrolysis products, in particular, the initial rising stages.

Insights into the pyrolysis kinetics of the three esters can be acquired with further model analyses. The following discussion will be based on relevant model simulations and analyses at 1500 K and 1.5 atm, the average conditions of those covered by the measurements [15]. Thin dash lines in Fig. 9 are simulated

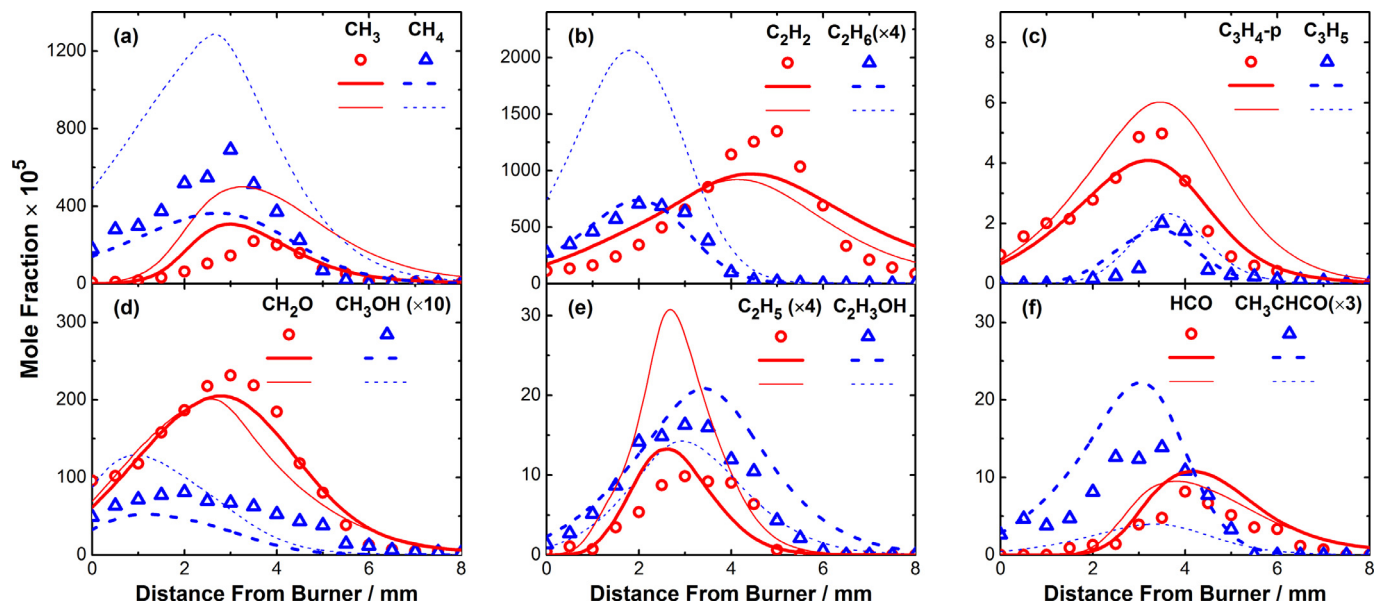


Fig. 7. Typical combustion intermediates in the EA flame. Symbols: experimental measurements; Thick lines: simulations with the current model; Thin lines: simulations with the Westbrook model [13].

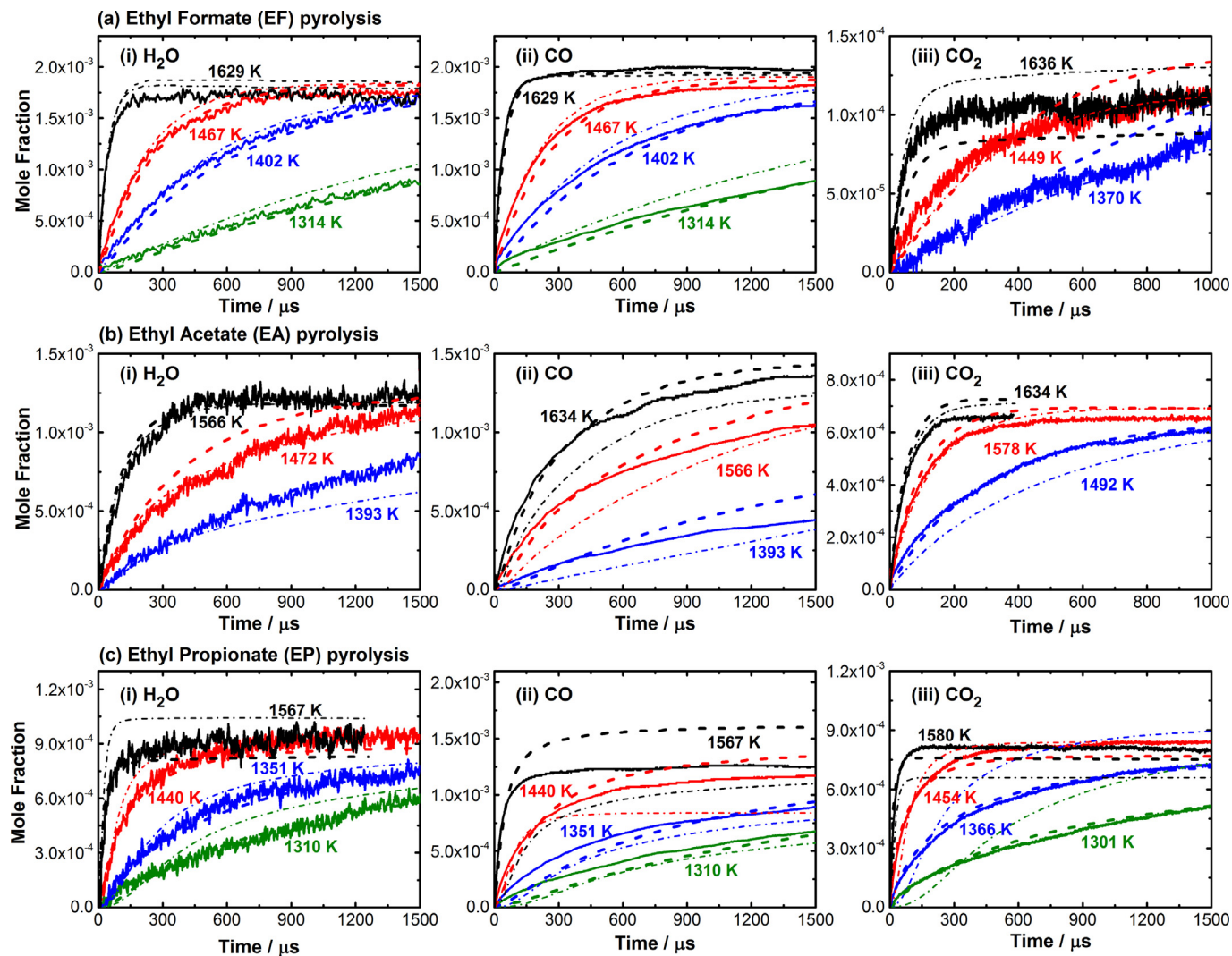


Fig. 8. Comparisons of the measured [15] and model predicting H_2O , CO and CO_2 concentration histories during the shock tube pyrolysis of (a) 2000 ppm ethyl formate (EF) in Ar, (b) 2000 ppm ethyl acetate (EA) in Ar and (c) 2000 ppm ethyl propionate (EP) in Ar. Solid lines: experimental measurements, thick dash lines: simulated results with the current model, thin dash-dot lines: simulated results with the model reported by Ren et al. [15].

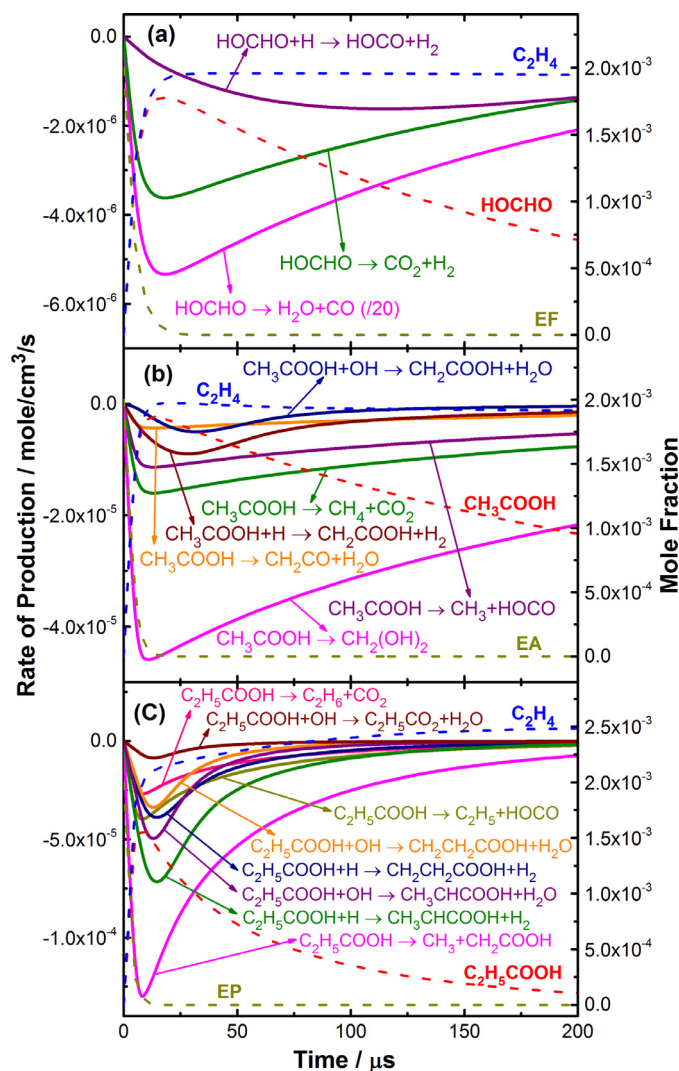


Fig. 9. ROP analysis (solid lines to left axes) for (a) formic acid (HOCHO) during EF pyrolysis, (b) acetic acid (CH₃COOH) during EA pyrolysis and (c) propionic acid (C₂H₅COOH) during EP pyrolysis. Dashed lines to right axes are simulated concentration time histories of fuels, C₂H₄ and corresponding acids. Conditions: 1500 K, 1.5 atm and 2000 ppm fuel in Ar.

concentrations for fuels, C₂H₄ and corresponding acids during the respective pyrolysis of EF, EA and EP with an initial fraction of 10⁻⁵, almost completely converting to C₂H₄ and acids, as can be proven by either the carbon atom balance or ROP analyses for fuels. The consumptions of acids cover much wider time range leading to the formation of numerous species. The time-varying ROP coefficients for acids consumption reactions are shown as the solid lines in Fig. 9. It can be noted that the dehydration channel predominantly accounts for the consumption of HOCHO, directly leading to formations of the measured products, CO and H₂O. For the decomposition of CH₃COOH, molecular reactions, including the dehydration [CH₃COOH → CH₂=C(OH)₂ → H₂O + CH₂CO] and the decarboxylation (CH₃COOH → CO₂ + CH₄), still dominate, but the C–C bond fission (CH₃COOH → CH₃ + HOCO) and hydrogen abstractions from the CH₃- sites (CH₃COOH + H/OH → ĊH₂COOH + H₂/H₂O) have non-negligible contributions. While in the case of C₂H₅COOH, only limited effects can be seen from the molecular reactions and the predominant consumption routes are shifted to bond fissions and hydrogen abstractions from the C₂H₅ moiety of C₂H₅COOH. A slow rise can be found in C₂H₄ concentration after the depletion

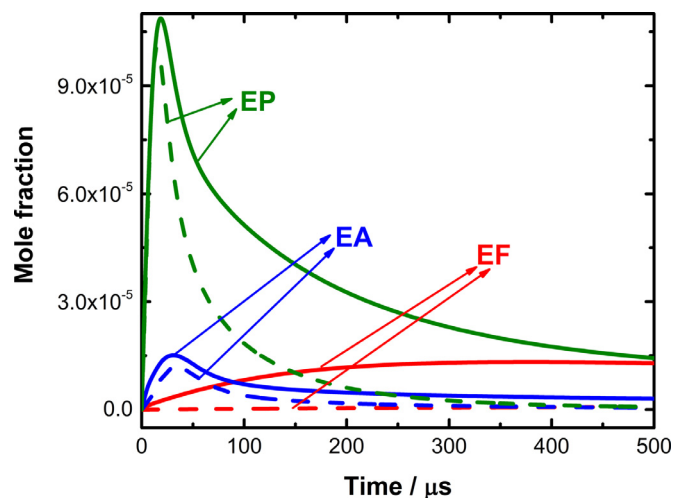


Fig. 10. Simulated concentration time histories of H atom (solid lines) and OH radical (dashed lines, values multiplied by 2) during the pyrolysis of three ethyl esters. Conditions: 1500 K, 1.5 atm and 2000 ppm fuel in Ar.

of EP, which is caused by the β-scission of the primary fuel radical (ĊH₂CH₂COOH) of C₂H₅COOH. The reaction schemes are quite various for the three acids whose consumptions have significant impacts on subsequent species formations. So different radical pools for the three pyrolysis mixtures could be expected.

The simulated concentration variations of H and OH during the pyrolysis of the three ethyl esters are presented in Fig. 10. Both radicals are of much higher concentrations during the pyrolysis of EP than those in cases of EF and EA, which is in line with the important role of hydrogen abstractions by H and OH in the consumption of C₂H₅COOH. A major source of H atom in all the three pyrolysis mixtures is the decomposition of HOCO radical (HOCO → H + CO₂), which can be yielded from either a direct bond fission of C₂H₅COOH (C₂H₅COOH → C₂H₅ + HOCO) or the β-scission of the primary radical (ĊH₂CH₂COOH) of C₂H₅COOH. It is noteworthy that OH radical is of a significant concentration in the EP pyrolysis mixture so that hydrogen abstractions by OH contribute much to the C₂H₅COOH consumption, as OH is usually of minimal levels under pyrolysis conditions. According to relevant ROP analyses, the OH radical in the EP pyrolysis mixture is originated from the hydroxyl (OH) moiety of C₂H₅COOH, which can be released through β-scissions of two radicals, ĊH₂COOH and CH₃ĊHCOOH. The former is a product of the principal C₂H₅COOH consumption channel C₂H₅COOH → CH₃ + ĊH₂COOH (see Fig. 9) and the latter is formed as hydrogen atoms are abstracted from the secondary carbon atom of C₂H₅COOH. It can be seen that H and OH can be respectively produced from the two C₂H₅COOH radicals, ĊH₂CH₂COOH and CH₃ĊHCOOH, thus automatically maintaining the reaction sequences initiated by hydrogen abstractions.

As an additional scrutiny, the current model was validated against measured ignition delay times [11,14,26] which could characterize the fuel reactivity. The tests covered a high temperature range of 1200–1800 K at different pressures between 1 and 12 atm, as demonstrated in Fig. 11. The current model can satisfactorily predict the ignition delay time measurements for all the three ethyl esters. To compare the reactivity of C₃–C₅ ethyl esters, model simulations were performed for stoichiometric mixtures of the three esters with the same argon/oxygen ratio (D) of 14 over the temperature range of 1200–1800 K at 1 atm. Under such investigated conditions, EA has shorter ignition delay times than those of EF and EP, as shown in Fig. S9. As mentioned above, CH₃CO and CH₃CO₂ radicals are easily formed in the consumption of EA and decompositions of these radicals can result in the higher CH₃ pro-

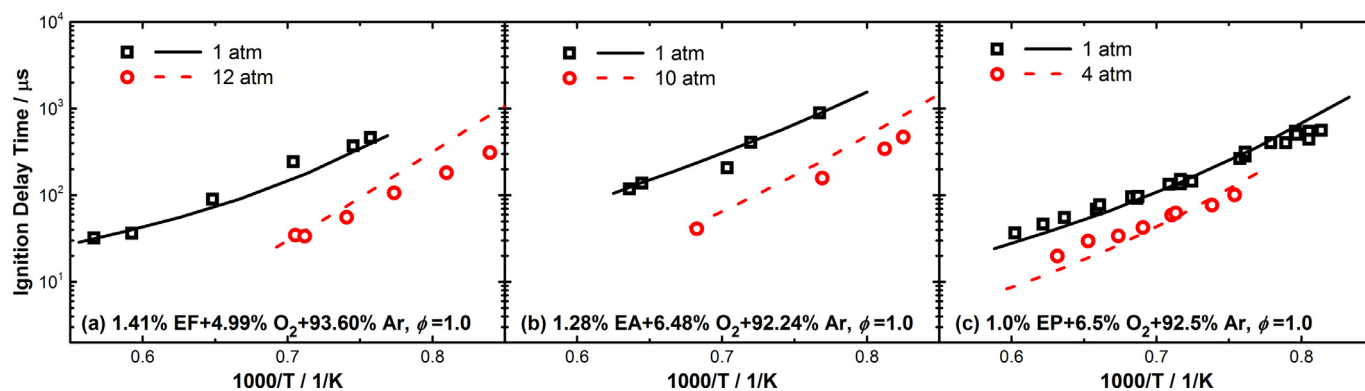


Fig. 11. Measured and simulated ignition delay times for ethyl esters/ O_2 /Ar mixtures over the temperature range of 1200–1800 K at different pressures. Symbols, measurements reported in [26] for EF, [14] for EA and [11] for EP; Lines: Simulations with the current model.

duction. Therefore H atoms can be removed by CH_3 through the reaction $\text{CH}_3 + \text{H} (+\text{M}) \rightarrow \text{CH}_4 (+\text{M})$, leading to the reduced reactivity of the EA mixture.

5. Conclusions

A better understanding of the high temperature combustion kinetics of C_3 – C_5 ethyl esters was pursued in the current work incorporating experimental, theoretical and modeling efforts. Chemical structures of fuel-rich flames of the three esters were respectively measured using a photoionization molecular-beam mass spectrometer (PI-MBMS), with focus on the “primary species”. Particularly, acids which can easily produce from unimolecular eliminations of corresponding ethyl esters were detected and quantified. A new kinetic model was developed, which shows improvements compared to previous models in predicting speciation measurements under flame and pyrolysis conditions. This can be attributed to the adoption of proper kinetic parameters derived theoretically from this work and literatures. And the inclusion of well-skipping reactions shows great potential in further improving model predictions for low-pressure flame structures. Intra-molecular eliminations and hydrogen abstractions by H atoms are competing pathways for fuel consumption under low-pressure premixed flame conditions where fall-off effects of the unimolecular reactions can never be ignored. During the shock tube pyrolysis, the conversions from fuels to C_2H_4 and acids accomplish in short periods, while subsequent decompositions of the acids accounts for the gradual accumulations of numerous species. H atoms and OH radicals are found abundant in the pyrolysis mixture of EP because they can yield from the dissociations of $\text{C}_2\text{H}_5\text{COOH}$ fuel radicals. Above-mentioned reaction schemes may also apply to long chain FAEE molecules though the relative significance of different reaction categories might vary. Rate coefficients for the dominant eliminations of larger ethyl esters would not differ much from those of the current small molecules, regarding the similar six-membered transition state rings. While more sites are available for hydrogen abstractions which could consequently play a more dominant role in fuel destructions.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2017.07.013.

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